

NOTES ON CCQME - EXPLICIT FORM

ALEXANDER SCHNELL

1. CCQME IN ENERGY BASIS

1.1. System-bath Hamiltonian as direct product. We start from a total system-plus-bath Hamiltonian

$$(1) \quad H_{\text{tot}} = H_S + H_B + s \otimes B.$$

For simplicity, here in the first step we assume a direct product of system-bath interaction Hamiltonian. The general case of multiple system and bath coupling operators is discussed below.

Performing the standard Born and Markov approximation, we arrive at the Redfield master equation

$$(2) \quad \partial_t \varrho(t) = -i [H_S, \varrho(t)] + \left([u \varrho(t), s] + \text{h.c.} \right) = -i [H_S, \varrho(t)] + \mathcal{R}[\varrho(t)],$$

where we define the Redfield superoperator $\mathcal{R}$ and Redfield jump operator

$$(3) \quad u = \int_0^\infty d\tau \tilde{s}(-\tau) C(\tau),$$

with bath correlation functions $C(\tau) = \langle \tilde{B}(\tau) B \rangle$ and $\tilde{A}(t)$ denotes an operator in the interaction picture. Practically, we can calculate these operators by introducing the eigenbasis of the system Hamiltonian $H_S |n\rangle = E_n |n\rangle$. This gives

$$(4) \quad u = \sum_{nm} s_{nm} W(\Delta_{nm}) |n\rangle \langle m|$$

where we have used energy differences $\Delta_{nm} = E_n - E_m$ and matrix elements $s_{nm} = \langle n | s | m \rangle$ and the Fourier-Laplace transform of the bath-correlation function

$$(5) \quad W(E) = \int_0^\infty d\tau e^{-\frac{i}{\hbar} E \tau} C(\tau).$$

Hence, we can explicitly denote the Redfield equation in the system's energy eigenbasis

$$(6) \quad \partial_t \varrho_{nm}(t) = -i \Delta_{nm} \varrho_{nm}(t) + \sum_{kl} \mathcal{R}_{nm,kl} \varrho_{kl}(t).$$

with

$$(7) \quad \mathcal{R}_{nm,kl} = \left(s_{lm}s_{nk}W(\Delta_{nk}) - \delta_{lm} \sum_q s_{nq}s_{qk}W(\Delta_{qk}) \right) + \left(s_{nk}s_{lm}W^*(\Delta_{ml}) - \delta_{kn} \sum_q s_{qm}s_{lq}W^*(\Delta_{ql}) \right).$$

With this convention, the CCQME is denoted as

$$(8) \quad \partial_t \varrho(t) = -i[H_S, \varrho(t)] + \mathcal{R}[(1 - \mathcal{Q})[\varrho(t)]].$$

Here, 1 is the identity superoperator and we introduce the superoperator $\mathcal{Q}$ which is given in the energy basis for $n \neq m$ via

$$(9) \quad \mathcal{Q}_{nm,kl} = \frac{\mathcal{R}_{nm,kl}}{i\Delta_{nm}}$$

and for $n = m$ and $k \neq l$ as

$$(10) \quad \mathcal{Q}_{nn,kl} = 0$$

and finally for $n = m$ and $k = l$

$$(11) \quad \mathcal{Q}_{nn,ll} = \sum_k |s_{nk}|^2 \left[V''(\Delta_{nk})\delta_{kl} - V''(\Delta_{kn})\delta_{ln} + W''(\Delta_{kn})\frac{F_{nl}}{f_n} \right]$$

Here we have written the function $W(E)$ in terms of its real- and imaginary part,

$$(12) \quad W(E) = W'(E) + iW''(E),$$

and similar for its derivative

$$(13) \quad V(E) = \frac{\partial W(E)}{\partial E} = V'(E) + iV''(E).$$

We also define for $n \neq l$

$$(14) \quad F_{nl} = |s_{nl}|^2 V'(\Delta_{nl})$$

and

$$(15) \quad F_{nn} = \sum_{k \neq n} |s_{nk}|^2 V'(\Delta_{kn}).$$

Finally we set

$$(16) \quad f_n = \sum_{k \neq n} |s_{kn}|^2 W'(\Delta_{kn}).$$

1.2. General system-bath Hamiltonian. In the general case, we start from a total system-plus-bath Hamiltonian

$$(17) \quad H_{\text{tot}} = H_S + H_B + \sum_{\alpha} s_{\alpha} \otimes B_{\alpha}.$$

The Redfield operator here reads

$$(18) \quad \mathcal{R}[\varrho(t)] = \sum_{\alpha} \left([u_{\alpha} \varrho(t), s_{\alpha}] + \text{h.c.} \right),$$

with

$$(19) \quad u_{\alpha} = \sum_{\beta, nm} s_{nm}^{\beta} W_{\alpha\beta}(\Delta_{nm}) |n\rangle\langle m|$$

and

$$(20) \quad W_{\alpha\beta}(E) = \int_0^{\infty} d\tau e^{-\frac{i}{\hbar} E \tau} C_{\alpha\beta}(\tau).$$

for the bath correlation functions $C_{\alpha\beta}(\tau) = \langle \tilde{B}_{\alpha}(\tau) B_{\beta} \rangle$.

Note that we now assume that the Bath operators B_{α} are defined such that the bath correlation functions are diagonal in the Bath operator index

$$(21) \quad C_{\alpha\beta}(\tau) = \delta_{\alpha\beta} C_{\alpha}(\tau).$$

Note that for a hopping-type term between system and bath this is not automatically fulfilled as we show below. Nevertheless a suitable definition of bath operators leads to a diagonal form.

Note that this form is required by the method of analytical continuation as presented in [PHYSICAL REVIEW E 88, 052127 (2013) <https://journals.aps.org/pre/pdf/10.1103/PhysRevE.88.052127>]. Maybe it is also okay in the old way that we did it but it is not directly justified, so let us check down below.

Hence, the Redfield tensor is

$$(22) \quad \begin{aligned} \mathcal{R}_{nm,kl} = & \sum_{\alpha} \left(s_{lm}^{\alpha} s_{nk}^{\alpha} W_{\alpha}(\Delta_{nk}) - \delta_{lm} \sum_q s_{nq}^{\alpha} s_{qk}^{\beta} W_{\alpha}(\Delta_{qk}) \right) \\ & + \sum_{\alpha} \left(s_{nk}^{\alpha} s_{lm}^{\alpha} W_{\alpha}^{*}(\Delta_{ml}) - \delta_{kn} \sum_q s_{qm}^{\alpha} s_{lq}^{\alpha} W_{\alpha}^{*}(\Delta_{ql}) \right). \end{aligned}$$

Here, again $\mathcal{Q}$ is given in the energy basis for $n \neq m$ via

$$(23) \quad \mathcal{Q}_{nm,kl} = \frac{\mathcal{R}_{nm,kl}}{i\Delta_{nm}}$$

and for $n = m$ and $k \neq l$ as

$$(24) \quad \mathcal{Q}_{nn,kl} = 0$$

and finally for $n = m$ and $k = l$

$$(25) \quad \mathcal{Q}_{nn,ll} = \sum_{\alpha,k} |s_{nk}^\alpha|^2 \left[V_\alpha''(\Delta_{nk}) \delta_{kl} - V_\alpha''(\Delta_{kn}) \delta_{ln} + W_\alpha''(\Delta_{kn}) \frac{F_{nl}}{f_n} \right]$$

Here we have for $n \neq l$

$$(26) \quad F_{nl} = \sum_{\alpha} |s_{nl}^\alpha|^2 V_\alpha'(\Delta_{nl})$$

and

$$(27) \quad F_{nn} = \sum_{\alpha, k \neq n} |s_{nk}^\alpha|^2 V_\alpha'(\Delta_{kn}),$$

as well as

$$(28) \quad f_n = \sum_{\alpha, k \neq n} |s_{kn}^\alpha|^2 W_\alpha'(\Delta_{kn}).$$